latter, as a gentle breeze will start something that will grow into a mighty avalanche, or a tiny spark will blow up a huge powder magazine. However, this hypothesis does not solve the difficulty completely. Because in every case known to us, while the amount of energy expanded in liberating a process is very small in comparison with the energy released, yet it does exist, even though it may have just a microscopic magnitude. Even the very gentlest breeze and very tiniest spark possess an energy above zero—and this is what matters here.

However, it is well known that there are some forces which produce a perceptible effect without any expenditure whatever of energy. These are what we may call “guiding forces,” such as, for instance, the resistance due to the rigidity of railroad rails which forces the wheels of a train to follow a pre-determined curve, without any expenditure of energy. An attempt might be made to ascribe a similar role to the mental forces in the guiding of physical processes along pre-determined paths in the human brain. But this, too, involves grave and insurmountable difficulties. For the modern science of brain physiology is based on the very premise that it is possible to achieve a satisfactory understanding of the laws of biological processes without postulating the intervention of any particular mental force. Such a hypothesis avoids also the theory of parallelism which, in contrast to the theory of interaction, assumes that mental and physical processes must, necessarily, run side by side, each according to its own laws, without interfering with each other. Of course, it still remains incomprehensible just how this reciprocal interdependence of two such fundamentally different occurrences is to be conceived, and whether it perhaps requires the assumption of some form of pre-established harmony. In this respect, the theory of parallelism, too, is hardly satisfactory.

And now, in order to get to the bottom of the matter, let us ask ourselves this question of basic significance: Just what do we know about mental processes? In what circumstances and in what sense may we speak of mental processes? Let us consider first where we come across mental processes in this world. We must take it for granted that members of the higher animal kingdom as well as human beings have emotions and sensations. But as we descend to the lower animals—where is the borderline where sensation ceases to exist? Has a worm any sensation of pain as it is crushed under our feet? And may plants be considered capable of some kind of sensation? There are botanists who are disposed to answer this question affirmatively. But such a theory can never be put to the test, let alone proved, and the wisest course seems to be not to venture any opinion in this regard. Along the entire ladder of evolution, from the lowest order of life up to Man, there is no point at which one can establish a discontinuity in the nature of mental processes.

It is nevertheless possible to specify a quite definite borderline, of decisive